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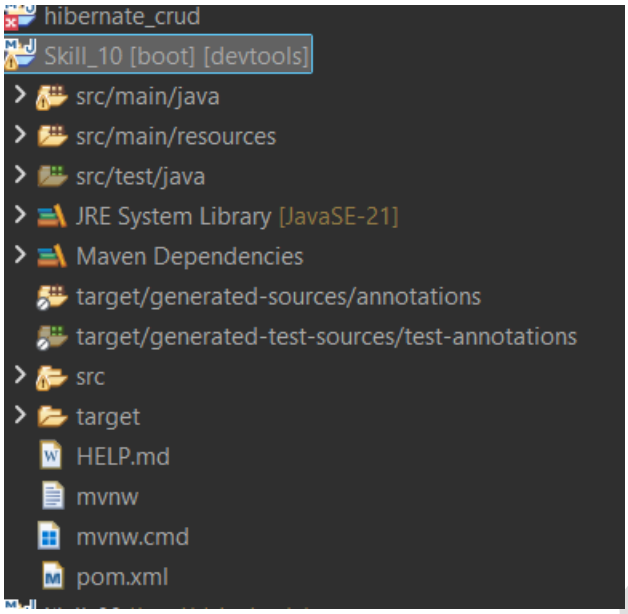
SECTION:59

Project Structure:



Student.java

```java
package klu.Spring_Boot_Demo;

import jakarta.persistence.*;

@Entity
@Table(name = "student_table")
public class Student {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;
    private String name;
    private String email;
    private String course;

    // Getters and Setters
    public Long getId() { return id; }
    public void setId(Long id) { this.id = id; }
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public String getEmail() { return email; }
    public void setEmail(String email) { this.email = email; }
    public String getCourse() { return course; }
    public void setCourse(String course) { this.course = course; }
}
```

StudentRepository.java

```java
package klu.Spring_Boot_Demo;

import org.springframework.data.jpa.repository.JpaRepository;

@Repository
public interface StudentRepository extends JpaRepository<Student, Long> {
}
```

StudentService.java

```java
package klu.Spring_Boot_Demo;

import org.springframework.beans.factory.annotation.Autowired;

@Service
public class StudentService {
    @Autowired
    private StudentRepository repository;

    public List<Student> getAllStudents() {
        return repository.findAll();
    }

    public Student saveStudent(Student student) {
        return repository.save(student);
    }

    public void deleteStudent(Long id) {
        repository.deleteById(id);
    }

    public Student getStudentById(Long id) {
        return repository.findById(id).orElse(null);
    }
}
```

StudentController.java

```java
package klu.Spring_Boot_Demo;

import org.springframework.beans.factory.annotation.Autowired;

@RestController
@RequestMapping("/students")
@CrossOrigin(origins = "http://localhost:5173") // Connects to your React app
public class StudentController {

    @Autowired
    private StudentService service;

    @GetMapping
    public List<Student> getAll() {
        return service.getAllStudents();
    }

    @PostMapping
    public ResponseEntity<Student> add(@RequestBody Student student) {
        return ResponseEntity.ok(service.saveStudent(student));
    }

    @PutMapping("/{id}")
    public ResponseEntity<Student> update(@PathVariable Long id, @RequestBody Student studentDetails) {
        Student student = service.getStudentById(id);
        if (student != null) {
            student.setName(studentDetails.getName());
            student.setEmail(studentDetails.getEmail());
            student.setCourse(studentDetails.getCourse());
            return ResponseEntity.ok(service.saveStudent(student));
        }
        return ResponseEntity.notFound().build();
    }

    @DeleteMapping("/{id}")
    public ResponseEntity<String> delete(@PathVariable Long id) {
        service.deleteStudent(id);
        return ResponseEntity.ok("Student deleted successfully");
    }
}
```

Application.properties:

```properties
spring.application.name=Spring_Boot_Demo

# H2 Database Configuration (In-Memory)
spring.datasource.url=jdbc:h2:mem:testdb
spring.datasource.driverClassName=org.h2.Driver
spring.datasource.username=sa
spring.datasource.password=
spring.jpa.database-platform=org.hibernate.dialect.H2Dialect

# Allows Hibernate to create the 'student_table' automatically
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true

# Enables the H2 Console so you can see your data at http://localhost:8080/h2-console
spring.h2.console.enabled=true
spring.h2.console.path=/h2-console

# Port Configuration
server.port=8080
```

SpringBootDemoApplication.java

```java
package klu.Spring_Boot_Demo;

import org.springframework.boot.SpringApplication;

@SpringBootApplication
public class SpringBootDemoApplication {
    public static void main(String[] args) {
        // This line tells Spring Boot to start the whole application
        SpringApplication.run(SpringBootDemoApplication.class, args);
    }
}
```

MainApplication.java

```
Skill_12/src/main/java/klu/Spring_Boot_Demo/StudentRepository.java
import java.util.Scanner;

public class MainApplication {

    public static void main(String[] args) {

        StudentManager manager = new StudentManager();
        Scanner sc = new Scanner(System.in);

        manager.displayStudents();

        System.out.println("Enter new student id:");
        int id = sc.nextInt();
        sc.nextLine();

        System.out.println("Enter student name:");
        String name = sc.nextLine();

        System.out.println("Enter course:");
        String course = sc.nextLine();

        manager.addStudent(new Student(id,name,course));

        System.out.println("Student List After Adding:");
        manager.displayStudents();

        System.out.println("Enter ID to delete:");
        int deleteId = sc.nextInt();
```

OUTPUT:

```java
import java.util.Scanner;

public class MainApplication {

    public static void main(String[] args) {

        StudentManager manager = new StudentManager();
        Scanner sc = new Scanner(System.in);

        manager.displayStudents();

        System.out.println("Enter new student id:");
        int id = sc.nextInt();
        sc.nextLine();

        System.out.println("Enter student name:");
        String name = sc.nextLine();

        System.out.println("Enter course:");
        String course = sc.nextLine();

        manager.addStudent(new Student(id, name, course));

        System.out.println("Student List After Adding:");
        manager.displayStudents();

        System.out.println("Enter ID to delete:");
        int deleteId = sc.nextInt();
```

```
Console X
Arjit   Python
Gaur    React
Pragya  SpringBoot
Anjali  AWS
Enter new student ID:
```

GitHub link: https://github.com/hemalatharachakonda/2400030211-Skill_experiments.git

2400030211